Chapter 1: Introduction

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1 Introduction

This chapter covers different objectives of experiments and introduces the concept of experimental design for data science.

Citation: V. R. Joseph (2026). *Experimental Design for Data Science and Engineering*. CRC Press.

2 Different Objectives of Experiments

```
library(support)

par(mar = c(5, 5, 4, 2))
par(mfrow = c(2, 2))

# Common parameters
x <- seq(0, 1, length = 301)
n <- 15

# 1. Approximate a rough function
y <- sin(10 * pi * x) / (1 + 64 * (x - 0.5)^2)
plot(x, y, type = "l", axes = FALSE, xlab = "", ylab = "", lwd = 3, col = 1,
     ylim = c(min(y) - 0.05, max(y)), main = "Approximate a Rough Function",
     cex.main = 2)
D <- ((1:n) - 0.5) / n  # Equi-spaced design
points(cbind(D, min(y) - 0.05), pch = 16, cex = 2, lwd = 3, col = "blue")

# 2. Approximate a smooth function
y <- 1 / (1 + 64 * (x - 0.5)^2)
plot(x, y, type = "l", axes = FALSE, xlab = "", ylab = "", lwd = 3, col = 1,
     ylim = c(min(y) - 0.05, max(y)), main = "Approximate a Smooth Function",
     cex.main = 2)
```

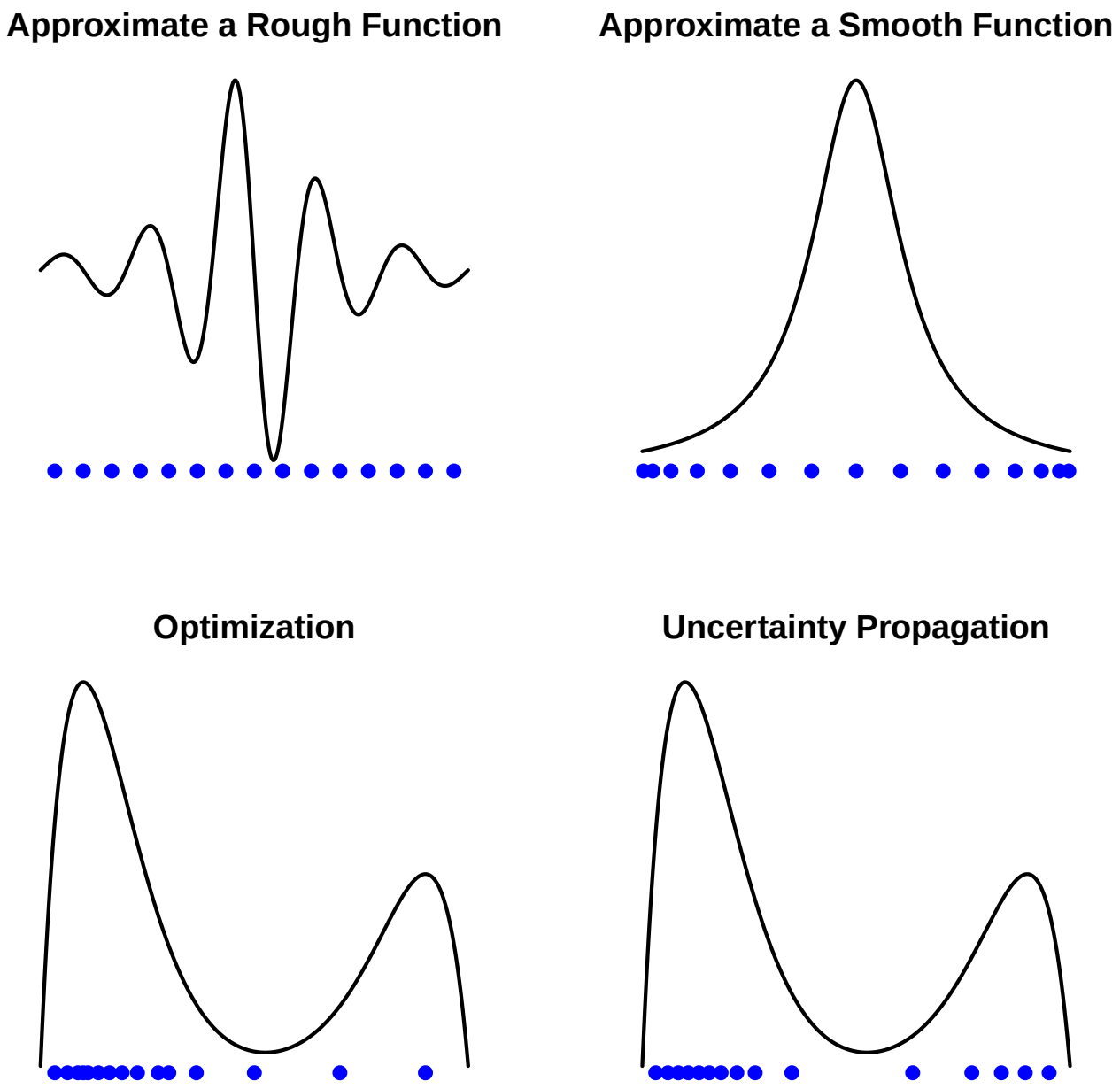
```r
D <- qbeta(((1:n) - 0.5) / n, 0.5, 0.5)  # Chebyshev-like design
points(cbind(D, min(y) - 0.05), pch = 16, cex = 2, lwd = 3, col = "blue")

# 3. Optimization
n1 <- n * 2/3
n2 <- n / 3
y <- 2/3 * dbeta(x, 2, 10) + 1/3 * dbeta(x, 10, 2)
plot(x, y, type = "l", axes = FALSE, xlab = "", ylab = "", lwd = 3, col = 1,
     ylim = c(min(y) - 0.05, max(y)), main = "Optimization", cex.main = 2)
D0 <- ((1:n2) - 0.5) / n2
D <- c(D0, qbeta(((1:n1) - 0.5) / n1, 2, 10))
points(cbind(D, min(y) - 0.05), pch = 16, cex = 2, lwd = 3, col = "blue")

# 4. Uncertainty propagation
y <- 2/3 * dbeta(x, 2, 10) + 1/3 * dbeta(x, 10, 2)
plot(x, y, type = "l", axes = FALSE, xlab = "", ylab = "", lwd = 3, col = 1,
     ylim = c(min(y) - 0.05, max(y)), main = "Uncertainty Propagation", cex.main = 2)
x_samp <- rbeta(1000 * n1, 2, 10)
x_samp <- c(x_samp, rbeta(1000 * n2, 10, 2))
a <- sp(n, 1, dist.samp = cbind(x_samp))
```

```
## Optimizing ...
## [>                                                                  ] 0 %[>
## Maximum iterations reached: done!
```

```r
D <- a$sp
points(cbind(D, min(y) - 0.05), pch = 16, cex = 2, lwd = 3, col = "blue")
```




```
par(mfrow = c(1, 1))
```

3 Linear Regression vs Gaussian Process

```
library(rkriging)

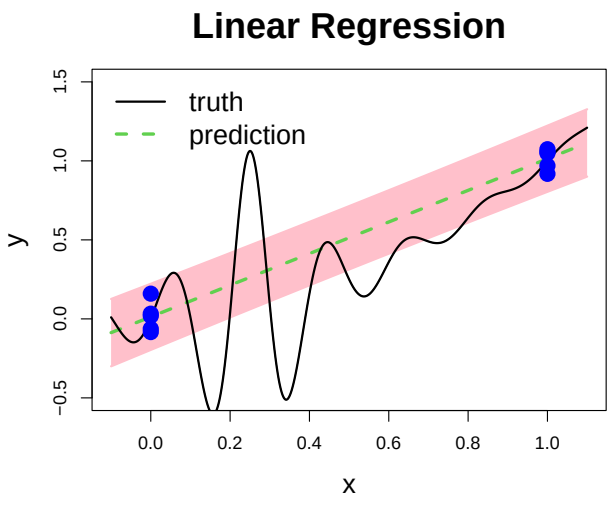
set.seed(1)
n <- 10
x <- rep(c(0, 1), c(n/2, n/2))
f <- function(x) sin(10 * pi * x) / (1 + 64 * (x - 0.25)^2) + x^2
y <- f(x) + 0.1 * rnorm(n)

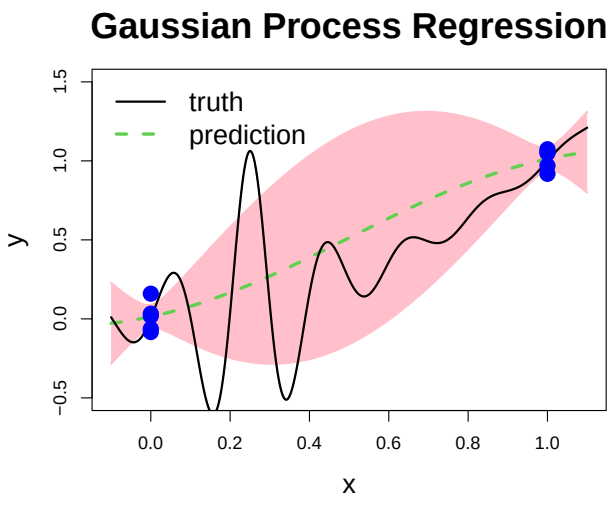
par(mfrow = c(1, 2))
l <- -0.1
```

```r
u <- 1.1
test <- seq(l, u, 0.001)
truey <- f(test)

# Linear Regression
plot(test, truey, type = "l", xlim = c(l, u), ylim = c(-0.5, 1.5),
     main = "Linear Regression", xlab = "x", ylab = "y", cex.main = 2, cex.lab = 1.5)
a.lm <- lm(y ~ x)
pred <- predict.lm(a.lm, newdata = data.frame(x = test), interval = "prediction")
low <- pred[, 2]
up <- pred[, 3]
polygon(c(test, rev(test)), c(low, rev(up)), col = "pink", border = NA, lwd = 2)
lines(test, low, lwd = 2, col = "pink")
lines(test, up, lwd = 2, col = "pink")
lines(test, pred[, 1], col = 3, lwd = 3, lty = 2)
lines(test, truey, xlim = c(l, u), lwd = 2)
points(x, y, pch = 16, col = "blue", cex = 2)
legend("topleft", legend = c("truth", "prediction"), lty = c(1, 2),
       col = c(1, 3), lwd = c(2, 3), bty = "n", cex = 1.5)

# Gaussian Process Regression
a.gp <- Fit.Kriging(x, y, interpolation = FALSE, fit = TRUE, model = "OK",
                    kernel.parameters = list(type = "Gaussian"))
pred <- Predict.Kriging(a.gp, test)
plot(test, truey, type = "l", xlim = c(l, u), ylim = c(-0.5, 1.5),
     main = "Gaussian Process Regression", xlab = "x", ylab = "y",
     cex.main = 2, cex.lab = 1.5)
low <- pred$mean - 2 * pred$sd
up <- pred$mean + 2 * pred$sd
polygon(c(test, rev(test)), c(low, rev(up)), col = "pink", border = NA, lwd = 2)
lines(test, low, lwd = 2, col = "pink")
lines(test, up, lwd = 2, col = "pink")
lines(test, pred$mean, col = 3, lwd = 3, lty = 2)
lines(test, truey, xlim = c(l, u), lwd = 2)
points(x, y, pch = 16, col = "blue", cex = 2)
legend("topleft", legend = c("truth", "prediction"), lty = c(1, 2),
       col = c(1, 3), lwd = c(2, 3), bty = "n", cex = 1.5)
```





```r
par(mfrow = c(1, 1))
```

4 Summary

This chapter introduced the fundamental concepts of experimental design:

1. **Space-filling designs** for approximating rough functions
2. **Non-uniform designs** (Chebyshev nodes) for smooth function approximation
3. **Adaptive designs** for optimization problems
4. **Support points** for uncertainty propagation

The comparison between linear regression and Gaussian process regression demonstrates the importance of choosing appropriate models for different types of data.